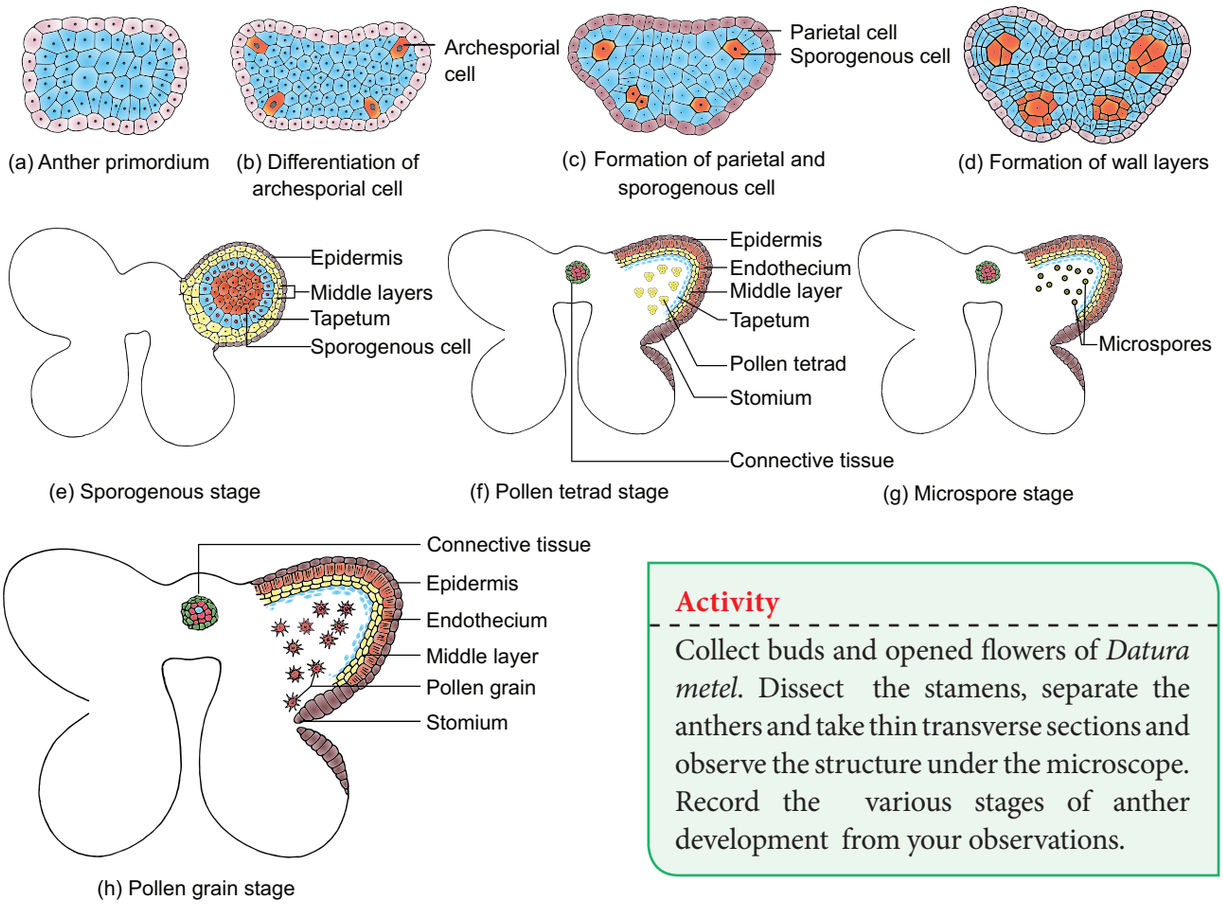


Figure 1.4 Stages in the development of anther

haploid microspores (microspore tetrad). The microspore tetrad may be arranged in a tetrahedral, decussate, linear, T shaped or isobilateral manner. Microspores soon separate from one another and remain free in the anther locule and develop into pollen grains. The stages in the development of microsporangia is given in Figure 1.4. In some plants, all the microspores in a microsporangium remain held together called **pollinium**. Example: *Calotropis*. Compound pollen grains are found in *Drosera* and *Drymis*.

T.S. of Mature anther

Transverse section of mature anther reveals the presence of anther cavity surrounded by an anther wall. It is bilobed, each lobe having 2 theca (dithecous). A typical anther is tetrasporangiate. The T.S. of Mature anther is given in Figure 1.5.

1. Anther wall

The mature anther wall consists of the following layers **a. Epidermis b. Endothecium c. Middle layers d. Tapetum.**

a. Epidermis: It is single layered and protective in function. The cells undergo repeated anticlinal divisions to cope up with the rapidly enlarging internal tissues.

b. Endothecium: It is generally a single layer of radially elongated cells found below the epidermis. The inner tangential wall develops bands (sometimes radial walls also) of α cellulose (sometimes also slightly lignified). The cells are **hygroscopic**. In the anthers of aquatic plants, saprophytes, cleistogamous flowers and extreme parasites endothecial differentiation is absent. The cells along the junction of the two sporangia of an anther lobe lack these thickenings. This region is called **stomium**. This region along with the hygroscopic nature of endothecium helps in the dehiscence of anther at maturity.

Activity

Collect buds and opened flowers of *Datura metel*. Dissect the stamens, separate the anthers and take thin transverse sections and observe the structure under the microscope. Record the various stages of anther development from your observations.

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